

Brutal Practice Paper B38 - Acids, Bases & Salts / Data Handling / Motion & Time / Algebraic Expressions

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A class tests 40 household liquids. 45% turn blue litmus red, 30% turn red litmus blue, and the rest change neither paper. How many liquids are neutral, and what do neutral liquids do to litmus? [\[Acids, Bases & Salts\]](#)
- (A) 22 are neutral; they cause no colour change
 - (B) 18 are neutral; they cause no colour change
 - (C) 8 are neutral; they cause no colour change
 - (D) 10 are neutral; they turn litmus blue
 - (E) 10 are neutral; they turn litmus red
 - (F) 10 are neutral; they cause no colour change
 - (G) 12 are neutral; they cause no colour change
2. A batsman's runs in 5 innings are 24, 38, 12, 40, 26. Find the mean, and say whether the mean rises or falls if his next innings is 0. [\[Data Handling\]](#)
- (A) mean 28; it rises
 - (B) mean 28; it stays the same
 - (C) mean 140; it falls
 - (D) mean 28; it falls
 - (E) mean 30; it falls
 - (F) mean 26; it falls
 - (G) mean 35; it falls
3. A car covers 240 km in 4 hours at a steady speed. Find its speed, and how far it would go in 7 hours at the same speed. [\[Motion & Time\]](#)
- (A) 80 km/h; 420 km
 - (B) 60 km/h; 480 km
 - (C) 60 km/h; 240 km
 - (D) 60 km/h; 360 km
 - (E) 60 km/h; 420 km
 - (F) 70 km/h; 420 km
4. In the expression $3x^2 - 5x + 7$, how many terms are there, and what is the constant term? [\[Algebraic Expressions\]](#)
- (A) 3 terms; constant 3
 - (B) 3 terms; constant 5
 - (C) 3 terms; constant 0
 - (D) 3 terms; constant 7
 - (E) 4 terms; constant 7
 - (F) 2 terms; constant 7
 - (G) 3 terms; constant -5

5. In a titration it takes 3 mL of base to neutralise every 2 mL of a certain acid. How much of this base is needed for 30 mL of the acid, and what is the reaction called? [\[Acids, Bases & Salts\]](#)

- (A) 45 mL of base; this is neutralisation
- (B) 45 mL of base; this is crystallisation
- (C) 60 mL of base; this is neutralisation
- (D) 15 mL of base; this is neutralisation
- (E) 20 mL of base; this is neutralisation
- (F) 30 mL of base; this is neutralisation
- (G) 45 mL of base; this is evaporation

6. Five students' heights in cm are 142, 130, 156, 138, 150. Find the median height and the range. [\[Data Handling\]](#)

- (A) median 150 cm; range 26 cm
- (B) median 142 cm; range 130 cm
- (C) median 142 cm; range 26 cm
- (D) median 138 cm; range 26 cm
- (E) median 142 cm; range 14 cm
- (F) median 143.2 cm; range 26 cm
- (G) median 142 cm; range 156 cm

7. A cyclist moves at a steady 18 km/h. Convert this speed into metres per second. [\[Motion & Time\]](#)

- (A) 6 m/s
- (B) 5 m/s
- (C) 10 m/s
- (D) 0.5 m/s
- (E) 1.8 m/s
- (F) 50 m/s
- (G) 18 m/s

8. In the term $-7xy$, what is the coefficient of xy , and is the term positive or negative? [\[Algebraic Expressions\]](#)

- (A) 7; negative
- (B) -1; negative
- (C) $-7y$; negative
- (D) -7 ; positive
- (E) xy ; negative
- (F) -7 ; the term is zero
- (G) -7 ; negative

9. Each antacid tablet neutralises 5 mL of stomach acid. A patient has 35 mL of excess acid. How many tablets fully neutralise it, and antacids are of what nature? [\[Acids, Bases & Salts\]](#)

- (A) 7 tablets; antacids are acidic
- (B) 8 tablets; antacids are basic
- (C) 7 tablets; antacids are basic
- (D) 6 tablets; antacids are basic
- (E) 35 tablets; antacids are basic
- (F) 7 tablets; antacids are neutral
- (G) 5 tablets; antacids are basic

- 10.** Shoe sizes sold in a shop one morning were 6, 7, 7, 8, 7, 9, 8, 7. Find the mode and how many times it occurs. [\[Data Handling\]](#)
- (A) mode 7.5; it occurs 4 times
 - (B) mode 8; it occurs 3 times
 - (C) mode 9; it occurs 1 time
 - (D) mode 6; it occurs 4 times
 - (E) mode 7; it occurs 3 times
 - (F) mode 7; it occurs 4 times
 - (G) mode 7; it occurs 5 times
- 11.** A pendulum takes 2 seconds to complete one full swing (its time period). How many complete oscillations does it make in one minute? [\[Motion & Time\]](#)
- (A) 2 oscillations
 - (B) 45 oscillations
 - (C) 120 oscillations
 - (D) 15 oscillations
 - (E) 30 oscillations
 - (F) 20 oscillations
- 12.** Add the like terms $5a + 3a + a$. What is the sum, and is the answer a single term? [\[Algebraic Expressions\]](#)
- (A) $9a$; yes, it is one term
 - (B) $9a^3$; yes, it is one term
 - (C) $8a$; yes, it is one term
 - (D) $15a$; yes, it is one term
 - (E) $9a$; no, it stays three terms
 - (F) $9a$; it becomes three terms
 - (G) 9; yes, it is one term
- 13.** Turmeric stays yellow in acids and neutral liquids but turns red in bases. Of 50 cloth stains tested, 0.36 of them turned red. How many stains were basic, and turmeric is which kind of indicator? [\[Acids, Bases & Salts\]](#)
- (A) 18 acids; turmeric is a natural indicator
 - (B) 18 stains; turmeric is a natural indicator
 - (C) 14 stains; turmeric is a natural indicator
 - (D) 32 stains; turmeric is a natural indicator
 - (E) 36 stains; turmeric is a natural indicator
 - (F) 18 stains; turmeric turns red in acids
 - (G) 18 stains; turmeric is a synthetic indicator
- 14.** Four numbers have an average of exactly 25. Three of the numbers are known to be 20, 30, and 28. Work out the fourth number and the sum of all four. [\[Data Handling\]](#)
- (A) fourth 22; total 78
 - (B) fourth 18; total 100
 - (C) fourth 25; total 100
 - (D) fourth 22; total 88
 - (E) fourth 20; total 100
 - (F) fourth 22; total 100

15. A pendulum makes 40 complete oscillations in 80 seconds. Find its time period, and say whether it is faster or slower than a pendulum with a 1-second period. [\[Motion & Time\]](#)

- (A) 3 s; slower
- (B) 1 s; slower
- (C) 2 s; slower
- (D) 0.5 s; slower
- (E) 2 s; same speed
- (F) 2 s; faster
- (G) 40 s; slower

16. Find the value of the expression $2x + 3$ when $x = 4$, and then when $x = -2$. [\[Algebraic Expressions\]](#)

- (A) 11 and 1
- (B) 11 and -4
- (C) 11 and -1
- (D) 11 and -7
- (E) 9 and -1
- (F) 8 and -1

17. An ant bite injects an acid into your skin. Rubbing baking soda (a base) on it gives relief. What process gives the relief, and what does baking soda solution do to red litmus? [\[Acids, Bases & Salts\]](#)

- (A) neutralisation; red litmus stays red
- (B) neutralisation; baking soda turns red litmus blue
- (C) crystallisation; baking soda turns red litmus blue
- (D) neutralisation; baking soda is an acid
- (E) condensation; baking soda turns red litmus blue
- (F) neutralisation; baking soda turns blue litmus red
- (G) evaporation; baking soda turns red litmus blue

18. A bag has 5 red, 3 blue, and 4 green marbles. One marble is drawn at random. Find the probability that it is blue, and the total number of marbles. [\[Data Handling\]](#)

- (A) $\frac{1}{4}$; total 12 marbles
- (B) $\frac{1}{2}$; total 12 marbles
- (C) $\frac{1}{12}$; total 12 marbles
- (D) $\frac{3}{4}$; total 12 marbles
- (E) $\frac{1}{4}$; total 11 marbles
- (F) $\frac{1}{3}$; total 12 marbles
- (G) $\frac{2}{4}$; total 12 marbles

19. A bus travels 60 km in 1 hour and then 90 km in the next 2 hours. Find its average speed for the whole journey. [\[Motion & Time\]](#)

- (A) 50 km
- (B) 45 km/h
- (C) 50 km/h
- (D) 75 km/h
- (E) 60 km/h
- (F) 30 km/h
- (G) 150 km/h

20. Subtract the expression $(2x + 5)$ from $(5x + 9)$. What is the result? [\[Algebraic Expressions\]](#)

- (A) $-3x - 4$
- (B) $3x + 4$
- (C) $3x + 14$
- (D) $7x + 14$
- (E) $3x + 5$
- (F) $-3x + 4$

21. A field's soil is too acidic. A farmer spreads quicklime (a base) at 4 kg per 100 square metres. For a field of 2500 square metres, how much lime is needed, and how does it change the soil? [\[Acids, Bases & Salts\]](#)

- (A) 100 kg of lime; the soil becomes more acidic
- (B) 100 kg of lime; the soil becomes less acidic
- (C) 40 kg of lime; the soil becomes less acidic
- (D) 250 kg of lime; the soil becomes less acidic
- (E) 25 kg of lime; the soil becomes less acidic
- (F) 1000 kg of lime; the soil becomes less acidic

22. A bar graph shows books read in a week: Mon 4, Tue 6, Wed 0, Thu 8, Fri 2. Find the mean per day, and on how many days were more books read than the mean? [\[Data Handling\]](#)

- (A) mean 4; 3 days above the mean
- (B) mean 4; 1 day above the mean
- (C) mean 20; 2 days above the mean
- (D) mean 5; 2 days above the mean
- (E) mean 4.5; 2 days above the mean
- (F) mean 4; 2 days above the mean

23. A car's odometer reads 45,260 km at the start and 45,500 km after a 4-hour drive. Find the distance covered and the average speed. [\[Motion & Time\]](#)

- (A) 90,760 km; 60 km/h
- (B) 260 km; 60 km/h
- (C) 240 km; 60 km/h
- (D) 240 km; 65 km/h
- (E) 200 km; 60 km/h
- (F) 240 km; 240 km/h
- (G) 240 km; 80 km/h

24. Which pair contains like terms: the pair $4x^2y$ and $-9x^2y$, or the pair $3ab$ and $3a^2b$? [\[Algebraic Expressions\]](#)

- (A) $3a^2b$ and $4x^2y$ are like terms
- (B) neither pair has like terms
- (C) $4x^2y$ and $-9x^2y$ are like terms
- (D) all four are like terms
- (E) $4x^2y$ and $3ab$ are like terms
- (F) both pairs are like terms
- (G) $3ab$ and $3a^2b$ are like terms

25. A factory's waste water turns blue litmus red. It is treated until neutral using 1 litre of base for every 4 litres of waste. For 600 litres of waste, how much base is needed, and what is the waste's nature? [\[Acids, Bases & Salts\]](#)

- (A) 120 L of base; the waste is acidic
- (B) 100 L of base; the waste is acidic
- (C) 150 L of base; the waste is acidic
- (D) 150 L of acid; the waste is basic
- (E) 150 L of base; the waste is basic
- (F) 2400 L of base; the waste is acidic
- (G) 75 L of base; the waste is acidic

26. Team A scored 35, 40, 45 and Team B scored 20, 60, 40 in three matches. Which team has the bigger range of scores, and by how much? [\[Data Handling\]](#)

- (A) Team B; bigger by 20
- (B) Team A; bigger by 30
- (C) Team B; bigger by 40
- (D) both have equal ranges
- (E) Team B; bigger by 30
- (F) Team A; bigger by 10
- (G) Team B; bigger by 50

27. On a distance-time graph, what does a horizontal straight line tell you about the object, and what does a sloping straight line tell you? [\[Motion & Time\]](#)

- (A) at rest; zero speed forever proven
- (B) moving fast; constant speed
- (C) at rest; changing speed
- (D) moving back; constant speed
- (E) speeding up; constant speed
- (F) at rest; constant speed
- (G) at rest; increasing speed

28. A number is x . 'Five more than twice the number' is written as an expression. What is the expression, and its value when $x = 6$? [\[Algebraic Expressions\]](#)

- (A) $x + 5$; value 11
- (B) $2x - 5$; value 7
- (C) $5x + 2$; value 32
- (D) $2x + 5$; value 17
- (E) $2x + 5$; value 12
- (F) $2x + 5$; value 11
- (G) $2(x + 5)$; value 22

- 29.** China rose indicator turns acids dark pink (magenta) and bases green. A student tests vinegar with it. What colour appears, and which acid does vinegar contain? [\[Acids, Bases & Salts\]](#)
- (A) yellow; vinegar has acetic acid
 - (B) green; vinegar has acetic acid
 - (C) green; vinegar has citric acid
 - (D) dark pink (magenta); vinegar has acetic acid
 - (E) dark pink (magenta); vinegar has citric acid
 - (F) dark pink (magenta); vinegar has lactic acid
 - (G) dark pink (magenta); vinegar has formic acid
- 30.** Three parcels weigh 1.2 kg, 800 g, and 1 kg. Find the mean weight in grams. [\[Data Handling\]](#)
- (A) 900 g
 - (B) 667 g
 - (C) 1000 g
 - (D) 1100 g
 - (E) 1000 kg
 - (F) 3000 g
- 31.** A train 200 m long moves at a steady 20 m/s. Find the time to completely pass a pole, and the time to completely cross a 300 m long bridge. [\[Motion & Time\]](#)
- (A) 25 s; 10 s
 - (B) 10 s; 25 s
 - (C) 10 s; 15 s
 - (D) 20 s; 25 s
 - (E) 4 s; 25 s
 - (F) 10 s; 35 s
 - (G) 10 s; 30 s
- 32.** A rectangle has length $(x + 3)$ and breadth x . Write its perimeter as an expression, and find its value when $x = 5$. [\[Algebraic Expressions\]](#)
- (A) $4x + 3$; value 23
 - (B) $4x + 6$; value 26
 - (C) $4x + 6$; value 20
 - (D) $4x + 12$; value 32
 - (E) $2x + 6$; value 16
 - (F) $2x + 3$; value 13
 - (G) $8x + 6$; value 46
- 33.** Hydrochloric acid reacts completely with sodium hydroxide solution. Name the two products formed in this reaction. [\[Acids, Bases & Salts\]](#)
- (A) calcium chloride and water
 - (B) sodium sulphate and water
 - (C) sodium chloride and oxygen
 - (D) hydrogen chloride and water
 - (E) sodium chloride only
 - (F) sodium chloride and water
 - (G) sodium hydroxide and water

- 34.** A fair die is rolled once. Find the probability of getting an even number, and the probability of getting a number greater than 4. [\[Data Handling\]](#)
- (A) $\frac{1}{2}$ and $\frac{1}{6}$
 - (B) $\frac{1}{3}$ and $\frac{1}{3}$
 - (C) $\frac{1}{6}$ and $\frac{1}{3}$
 - (D) $\frac{1}{2}$ and $\frac{1}{4}$
 - (E) $\frac{1}{2}$ and $\frac{1}{2}$
 - (F) $\frac{1}{2}$ and $\frac{1}{3}$
 - (G) $\frac{1}{3}$ and $\frac{1}{2}$
- 35.** A runner finishes 1500 m in 5 minutes. Find the speed in metres per minute and in metres per second. [\[Motion & Time\]](#)
- (A) 250 m/min; 5 m/s
 - (B) 300 m/min; 5 m/s
 - (C) 1500 m/min; 5 m/s
 - (D) 300 m/min; 3 m/s
 - (E) 300 m/min; 50 m/s
 - (F) 300 m/min; 6 m/s
- 36.** Add the three expressions $3x + 2y$, $x - y$, and $2x + 4y$. What is the sum? [\[Algebraic Expressions\]](#)
- (A) $6x + 6y$
 - (B) $5x + 5y$
 - (C) $6x + 4y$
 - (D) $6x + 5y$
 - (E) $6x - 5y$
 - (F) $4x + 5y$
 - (G) $6x + 7y$
- 37.** Curd contains lactic acid. A dairy tests 80 milk samples and $\frac{5}{8}$ of them had gone sour. How many samples soured, and what happens to acidity as milk sours? [\[Acids, Bases & Salts\]](#)
- (A) 50 samples; acidity increases as it sours
 - (B) 10 samples; acidity increases as it sours
 - (C) 30 samples; acidity increases as it sours
 - (D) 48 samples; acidity increases as it sours
 - (E) 50 samples; acidity decreases as it sours
 - (F) 64 samples; acidity increases as it sours
 - (G) 50 samples; it becomes basic
- 38.** The mean of 6 numbers is 15. A seventh number, 22, is then added to the set. Find the new mean of all 7 numbers. [\[Data Handling\]](#)
- (A) 16.2
 - (B) 16
 - (C) 15.5
 - (D) 18.5
 - (E) 17
 - (F) 14

39. Car A covers 120 km in 1.5 hours and Car B covers 100 km in 1.25 hours. Which car is faster? [\[Motion & Time\]](#)
- (A) both travel at 60 km/h
 - (B) Car A 80, Car B 75; A faster
 - (C) Car A at 90 km/h is faster
 - (D) both have equal speed of 80 km/h
 - (E) Car A is faster
 - (F) Car B is faster
 - (G) Car B at 80, A at 75; B faster
40. Simplify $2x^2 + 3x^2 - x^2$. What is the result, and what is its degree (highest power)? [\[Algebraic Expressions\]](#)
- (A) $6x^2$; degree 2
 - (B) $4x^6$; degree 6
 - (C) $4x^2$; degree 1
 - (D) $5x^2$; degree 2
 - (E) $4x^2$; degree 2
 - (F) $4x^2$; degree 4
 - (G) $4x^3$; degree 3
41. Phenolphthalein is colourless in acid and pink in base. It takes 2 drops of acid to remove the pink from 5 drops of base. How many drops of acid clear a flask equal to 40 drops of base, and what is the flask then? [\[Acids, Bases & Salts\]](#)
- (A) 20 drops; the solution becomes neutral
 - (B) 8 drops; the solution becomes neutral
 - (C) 16 drops; pink because basic
 - (D) 16 drops; the solution becomes neutral
 - (E) 16 drops; the solution becomes acidic
 - (F) 100 drops; the solution becomes neutral
 - (G) 16 drops; the solution stays pink
42. A test gave marks 8, 5, 8, 9, 5, 8, 10. Find the mode and the median of these marks. [\[Data Handling\]](#)
- (A) mode 5, median 8
 - (B) mode 8, median 8
 - (C) mode 8, median 7
 - (D) mode 8, median 5
 - (E) mode 8, median 8.5
 - (F) mode 8, median 9
 - (G) mode 9, median 8
43. Two pendulums hang side by side, one longer and one shorter. Which one has the larger time period (swings more slowly)? [\[Motion & Time\]](#)
- (A) both swing equally
 - (B) the shorter pendulum swings more slowly
 - (C) the shorter one has the larger period
 - (D) the longer pendulum swings more slowly (larger period)
 - (E) the heavier bob swings more slowly
 - (F) the length does not matter
 - (G) the longer pendulum swings faster

44. Evaluate the expression $x^2 - 2x$ when $x = -3$. [\[Algebraic Expressions\]](#)

- (A) -15
- (B) 6
- (C) 9
- (D) 3
- (E) -3
- (F) 12
- (G) 15

45. Rain is normally slightly acidic, but pollution can make it strongly acidic. Of 120 rainy days in a city, 0.25 had strongly acidic rain that harmed plants. How many such days were there, and which gas mainly causes acid rain? [\[Acids, Bases & Salts\]](#)

- (A) 30 days; oxygen
- (B) 30 days; sulphur dioxide
- (C) 60 days; sulphur dioxide
- (D) 30 days; carbon monoxide
- (E) 25 days; sulphur dioxide
- (F) 48 days; sulphur dioxide

46. The daily high temperatures for a week (in deg C) were 30, 32, 28, 34, 31, 29, 33. Find the mean high, rounded to the nearest whole number. [\[Data Handling\]](#)

- (A) 30 deg C
- (B) 33 deg C
- (C) 31 deg C
- (D) 32 deg C
- (E) 29 deg C
- (F) 217 deg C
- (G) 31.5 deg C

47. Which is the fastest speed: 54 km/h, 12 m/s, or 600 m/min? [\[Motion & Time\]](#)

- (A) all three are equal
- (B) 600 m/min (equal to 12 m/s)
- (C) 12 m/s
- (D) 54 km/h (equal to 15 m/s)
- (E) 54 km/h equal to 5 m/s
- (F) 600 m/min
- (G) 12 m/s (equal to 50 km/h)

48. Which expression is a common factor of both $6ab$ and $9a$? [\[Algebraic Expressions\]](#)

- (A) $3ab$
- (B) ab
- (C) $6a$
- (D) $3a$
- (E) $18a$
- (F) $9ab$

49. Toothpaste is basic and neutralises the acid that bacteria make in the mouth. If brushing neutralises 6 units of acid twice a day, how much acid is neutralised in one week, and what does toothpaste do to red litmus? [\[Acids, Bases & Salts\]](#)

- (A) 84 units; no change to litmus
- (B) 84 units; red litmus stays red
- (C) 84 units; red litmus turns blue
- (D) 98 units; red litmus turns blue
- (E) 12 units; red litmus turns blue
- (F) 42 units; red litmus turns blue
- (G) 84 units; blue litmus turns red

50. A spinner has 8 equal parts: 3 red, 2 blue, 2 yellow, and 1 green. Find the probability that it does NOT land on red. [\[Data Handling\]](#)

- (A) $\frac{3}{8}$
- (B) $\frac{5}{8}$
- (C) $\frac{1}{8}$
- (D) $\frac{3}{5}$
- (E) $\frac{5}{3}$
- (F) $\frac{6}{8}$

51. A journey is 180 km driven at a steady 60 km/h, with a 30-minute rest stop on the way. Find the total time for the journey. [\[Motion & Time\]](#)

- (A) 4 hours
- (B) 3 hours 30 minutes
- (C) 3 hours 50 minutes
- (D) 3.3 hours
- (E) 3 hours
- (F) 2 hours 30 minutes
- (G) 3 hours 18 minutes

52. In the term $4x^2y^3$, what is the numerical coefficient, and what is the sum of the powers of the letters?

[\[Algebraic Expressions\]](#)

- (A) 4; total power 3
- (B) 8; total power 5
- (C) 12; total power 5
- (D) 4; total power 2
- (E) 4; total power 5
- (F) 4; total power 6

53. Common salt solution is tested with both red and blue litmus papers. What is seen, and from what kinds of substances is a salt usually made? [\[Acids, Bases & Salts\]](#)

- (A) no colour change; a salt forms from a base only
- (B) blue litmus turns red; a salt forms from an acid and a base
- (C) no colour change; a salt forms from two bases
- (D) no colour change; a salt forms from an acid and a base
- (E) red litmus turns blue; a salt forms from an acid and a base
- (F) no colour change; a salt forms from two acids

54. In a frequency table the value 10 occurs 3 times, 20 occurs 5 times, and 30 occurs 2 times. Find the mean of all these values. [\[Data Handling\]](#)

- (A) 190
- (B) 19
- (C) 63.3
- (D) 20
- (E) 18
- (F) 17
- (G) 21

55. A train travels 360 km at a steady speed, taking 4.5 hours. Find its speed, and the time it needs to cover 480 km at that speed. [\[Motion & Time\]](#)

- (A) 80 km/h; 4 hours
- (B) 60 km/h; 6 hours
- (C) 80 km/h; 6.5 hours
- (D) 90 km/h; 6 hours
- (E) 72 km/h; 6 hours
- (F) 80 km/h; 6 hours

56. Simplify the expression $7m - 3m - m$. [\[Algebraic Expressions\]](#)

- (A) $-3m$
- (B) $6m$
- (C) $5m$
- (D) $11m$
- (E) $4m$
- (F) $3m^2$
- (G) $3m$

57. 30 mL of an acid needs 30 mL of an equally strong base to become neutral. If only 18 mL of the base is added, what is the mixture, and how much more base is needed? [\[Acids, Bases & Salts\]](#)

- (A) neutral; no more base is needed
- (B) still acidic; 18 mL more base is needed
- (C) still acidic; 30 mL more base is needed
- (D) still acidic; 48 mL more base is needed
- (E) still acidic; 12 mL more base is needed
- (F) basic; 12 mL more acid is needed
- (G) basic; 12 mL more base is needed

58. The lowest temperatures recorded on five days were -3 , 5 , -8 , 2 , and 4 deg C. Find the range of these temperatures. [\[Data Handling\]](#)

- (A) -13 deg C
- (B) 8 deg C
- (C) 11 deg C
- (D) 3 deg C
- (E) 10 deg C
- (F) 16 deg C
- (G) 13 deg C

59. A pendulum completes 150 oscillations in 5 minutes. Find its time period in seconds. [\[Motion & Time\]](#)

- (A) 0.5 s
- (B) 2 s
- (C) 3 s
- (D) 1 s
- (E) 150 s
- (F) 30 s
- (G) 2.5 s

60. Riya is x years old. Her father is 4 years more than three times her age. Write her father's age as an expression, and find it when $x = 10$. [\[Algebraic Expressions\]](#)

- (A) $3x + 4$; 24 years
- (B) $3x + 4$; 34 years
- (C) $x + 4$; 14 years
- (D) $3x + 4$; 30 years
- (E) $3(x + 4)$; 42 years
- (F) $3x - 4$; 26 years
- (G) $4x + 3$; 43 years

61. A wasp sting is basic, unlike a bee sting which is acidic. Which mild household substance would best relieve a wasp sting, and why? [\[Acids, Bases & Salts\]](#)

- (A) baking soda, because a wasp sting is basic
- (B) milk of magnesia, because the sting is basic
- (C) vinegar, because a wasp sting is basic
- (D) lime water, because a wasp sting is basic
- (E) soap solution, because the sting is basic
- (F) baking soda, because the sting is acidic
- (G) vinegar, because a wasp sting is acidic

62. Six numbers are 12, 7, 15, 9, 20, and 11. Find the median. [\[Data Handling\]](#)

- (A) 9
- (B) 12
- (C) 10
- (D) 13
- (E) 13.5
- (F) 11.5

63. A snail crawls at a steady 0.5 cm per second. How far does it travel in 4 minutes, in centimetres? [\[Motion & Time\]](#)

- (A) 2 cm
- (B) 120 m
- (C) 60 cm
- (D) 240 cm
- (E) 48 cm
- (F) 12 cm
- (G) 120 cm

64. How many terms are in the expression $5xy - 7x + 2y - 9$, and which is the constant term? [\[Algebraic](#)

[Expressions\]](#)

- (A) 4 terms; constant -7
- (B) 4 terms; constant 9
- (C) 4 terms; no constant term
- (D) 5 terms; constant -9
- (E) 4 terms; constant 2
- (F) 4 terms; constant -9
- (G) 3 terms; constant -9

65. An acid sample needs 200 mL of base in total to be neutralised. The base is added 50 mL at a time. How many additions are needed, and what is the reaction called? [\[Acids, Bases & Salts\]](#)

- (A) 8 additions; neutralisation
- (B) 4 additions; evaporation
- (C) 3 additions; neutralisation
- (D) 250 additions; neutralisation
- (E) 4 additions; crystallisation
- (F) 4 additions; neutralisation
- (G) 5 additions; neutralisation

66. A class of 20 students scored a mean of 14 marks in a quiz. Find the total marks scored by the whole class. [\[Data Handling\]](#)

- (A) 260
- (B) 14
- (C) 200
- (D) 34
- (E) 280
- (F) 2800
- (G) 140

67. Which device is best for measuring very small time intervals accurately, as used in modern watches?

[\[Motion & Time\]](#)

- (A) a pendulum clock
- (B) a sundial
- (C) a quartz clock
- (D) a sand clock
- (E) a speedometer
- (F) an odometer

68. Collect the like terms in $4a + 3b - 2a + 5b$. [\[Algebraic Expressions\]](#)

- (A) $2a - 8b$
- (B) $6a + 2b$
- (C) $8a + 2b$
- (D) $2a + 15b$
- (E) $6a + 8b$
- (F) $2a + 8b$

- 69.** Lemon juice contains citric acid. A drink mixes lemon juice and water in the ratio 3 to 5, making 240 mL in all. How much lemon juice is in it, and is the drink acidic or basic? [\[Acids, Bases & Salts\]](#)
- (A) 90 mL of lemon juice; the drink is salty-neutral
 - (B) 120 mL of lemon juice; the drink is acidic
 - (C) 80 mL of lemon juice; the drink is acidic
 - (D) 90 mL of lemon juice; the drink is neutral
 - (E) 90 mL of lemon juice; the drink is acidic
 - (F) 150 mL of lemon juice; the drink is acidic
 - (G) 90 mL of lemon juice; the drink is basic
- 70.** In a box of 30 pens, 6 are faulty. A pen is picked at random. Find the probability that it is a good (non-faulty) pen. [\[Data Handling\]](#)
- (A) $\frac{4}{5}$
 - (B) $\frac{3}{5}$
 - (C) $\frac{4}{6}$
 - (D) $\frac{24}{5}$
 - (E) $\frac{6}{30}$
 - (F) $\frac{1}{5}$
- 71.** In a moving vehicle, what does the speedometer show and what does the odometer show? [\[Motion & Time\]](#)
- (A) distance travelled; time
 - (B) speed; fuel left
 - (C) speed; distance travelled
 - (D) speed; time
 - (E) fuel left; distance travelled
 - (F) distance travelled; speed
- 72.** Find the value of the expression $(3x)/2 + 1$ when $x = 4$. [\[Algebraic Expressions\]](#)
- (A) 4
 - (B) 6
 - (C) 7
 - (D) 7.5
 - (E) 5
 - (F) 9
 - (G) 13
- 73.** Milk of magnesia is a base taken for acidity. If one dose neutralises 8 mL of stomach acid and a patient has 56 mL of excess acid, how many doses are needed, and what colour does it give with turmeric? [\[Acids, Bases & Salts\]](#)
- (A) 6 doses; turmeric turns red
 - (B) 8 doses; turmeric turns red
 - (C) 7 doses; turmeric turns red
 - (D) 7 doses; turmeric turns green
 - (E) 9 doses; turmeric turns red
 - (F) 7 doses; turmeric turns yellow

- 74.** Group X has 5 members with a mean of 12, and Group Y has 5 members with a mean of 18. Find the combined mean of all 10 members. [\[Data Handling\]](#)
- (A) 18
 - (B) 14
 - (C) 16
 - (D) 15
 - (E) 12
 - (F) 30
 - (G) 75
- 75.** You walk to school, 1.5 km in 30 minutes, and return by the same route in 20 minutes. Find the total distance and the average speed in km/h. [\[Motion & Time\]](#)
- (A) 3 km; 6 km/h
 - (B) 6 km; 3.6 km/h
 - (C) 3 km; 3 km/h
 - (D) 3 km; 3.6 km/h
 - (E) 1.5 km; 3.6 km/h
 - (F) 3 km; 2.5 km/h
- 76.** Write the product 3 multiplied by x multiplied by x using powers. [\[Algebraic Expressions\]](#)
- (A) $3x^3$
 - (B) x^3
 - (C) $6x$
 - (D) $3x^2$
 - (E) $9x^2$
 - (F) $3x$
 - (G) x^2
- 77.** Soap solution is basic. It is tested with a red litmus paper and a blue litmus paper. What happens to each paper? [\[Acids, Bases & Salts\]](#)
- (A) blue litmus turns red; red litmus stays red
 - (B) red litmus turns blue; blue litmus stays blue
 - (C) blue litmus turns red; red litmus turns blue
 - (D) red litmus turns blue; blue litmus turns red
 - (E) red litmus stays red; blue litmus turns red
 - (F) neither paper changes
- 78.** Class A has 10 students with a mean of 20, and Class B has 5 students with a mean of 26. Find the combined mean of all 15 students. [\[Data Handling\]](#)
- (A) 46
 - (B) 22
 - (C) 330
 - (D) 25
 - (E) 21
 - (F) 23
 - (G) 24

79. A car covers equal distances in equal intervals of time throughout its journey. What kind of motion is this called? [\[Motion & Time\]](#)

- (A) random motion
- (B) slow motion
- (C) uniform motion
- (D) periodic motion
- (E) circular motion
- (F) non-uniform motion

80. Subtract the expression $(4x - 7)$ from $(x - 2)$. What is the result? [\[Algebraic Expressions\]](#)

- (A) $-3x + 5$
- (B) $3x + 5$
- (C) $5x - 9$
- (D) $3x - 5$
- (E) $-3x + 9$
- (F) $-5x + 5$

81. Vinegar (an acid) is mixed with lemon juice (also an acid). What is the nature of the mixture, and what will it do to litmus? [\[Acids, Bases & Salts\]](#)

- (A) acidic; it turns turmeric red
- (B) neutral; it causes no change
- (C) basic; it turns red litmus blue
- (D) basic; it causes no change
- (E) neutral; it turns blue litmus red
- (F) acidic; it turns red litmus blue
- (G) acidic; it turns blue litmus red

82. A bar graph shows monthly sales: Jan 200, Feb 250, Mar 300, Apr 250. Find the total sales, and what percent of the total was March's sales? [\[Data Handling\]](#)

- (A) 1000; 33%
- (B) 1000; 30%
- (C) 1000; 3%
- (D) 1100; 30%
- (E) 1000; 25%
- (F) 900; 30%
- (G) 1000; 300%

83. On a distance-time graph, what does a steeper line tell you about an object's motion, and why? [\[Motion & Time\]](#)

- (A) it moves faster; in the same time it covers more distance
- (B) it moves slower; in the same time it covers more distance
- (C) it moves backward; it covers more distance
- (D) it moves slower; it covers less distance
- (E) it is at rest; it covers no distance
- (F) it moves faster; it covers the same distance
- (G) it moves faster; in the same time it covers less distance

- 84.** A number pattern has n th term $2n + 1$. Find the 10th term and the difference between the 10th and 9th terms. [\[Algebraic Expressions\]](#)
- (A) 21; difference 3
 - (B) 20; difference 2
 - (C) 23; difference 2
 - (D) 21; difference 21
 - (E) 19; difference 2
 - (F) 21; difference 1
 - (G) 21; difference 2
- 85.** Six liquids are tested: lemon juice, soap solution, vinegar, baking soda solution, curd, and lime water. How many of these six are acidic? [\[Acids, Bases & Salts\]](#)
- (A) 1 is acidic
 - (B) 3 are basic
 - (C) 2 are acidic
 - (D) 5 are acidic
 - (E) 4 are acidic
 - (F) 3 are acidic
- 86.** A data set is 4, 5, 6, 7, 8, with each value appearing exactly once. What is the mode of this data? [\[Data Handling\]](#)
- (A) the mode is 5
 - (B) the mode is 8
 - (C) there is no mode
 - (D) the mode is 30
 - (E) the mode is 4
 - (F) the mode is the mean, 6
- 87.** A car covers 1.2 km in 48 seconds at a steady speed. Find its speed in metres per second and in kilometres per hour. [\[Motion & Time\]](#)
- (A) 25 m/s; 72 km/h
 - (B) 20 m/s; 90 km/h
 - (C) 30 m/s; 90 km/h
 - (D) 2.5 m/s; 90 km/h
 - (E) 25 m/s; 90 km/h
 - (F) 25 m/s; 100 km/h
- 88.** A triangle has sides x , $x + 2$, and $2x - 1$. Write its perimeter as an expression, and find it when $x = 4$. [\[Algebraic Expressions\]](#)
- (A) $4x + 1$; value 16
 - (B) $4x + 1$; value 18
 - (C) $3x + 1$; value 13
 - (D) $4x + 1$; value 17
 - (E) $4x - 1$; value 15
 - (F) $4x + 3$; value 19
 - (G) $2x + 1$; value 9

89. When an acid and a base react they form salt and water and give out some heat. In 5 identical trials each releasing 12 units of heat, what is the total heat, and what are the products? [\[Acids, Bases & Salts\]](#)

- (A) 48 units; salt and water
- (B) 60 units; salt and oxygen
- (C) 60 units; only salt
- (D) 60 units; salt and water
- (E) 60 units; acid and base
- (F) 72 units; salt and water
- (G) 17 units; salt and water

90. The data set 2, 4, 6, 8 has a median of 5. If the value 8 is changed to 100, what is the new median?

[\[Data Handling\]](#)

- (A) it becomes 28
- (B) it becomes 5.5
- (C) it becomes 4
- (D) still 5
- (E) it becomes 53
- (F) it becomes 6
- (G) it becomes 50

91. Two friends stand 100 m apart and walk straight toward each other, each at 2 m/s. After how many seconds do they meet? [\[Motion & Time\]](#)

- (A) 50 s
- (B) 25 s
- (C) 25 m
- (D) 12.5 s
- (E) 20 s
- (F) 40 s
- (G) 100 s

92. Simplify the expression $3x + 2y - x + 5 - y$. After simplifying, how many terms remain? [\[Algebraic Expressions\]](#)

- (A) 3 terms ($2x + y + 5$)
- (B) 5 terms
- (C) 4 terms
- (D) 6 terms
- (E) 3 terms ($2x + 3y + 5$)
- (F) 2 terms
- (G) 1 term

93. Baking powder is baking soda mixed with a mild edible acid. When wet, the two react and release a gas that makes a cake rise. Name the gas, and name the kind of reaction. [\[Acids, Bases & Salts\]](#)

- (A) carbon dioxide; evaporation
- (B) oxygen; neutralisation
- (C) carbon dioxide; condensation
- (D) carbon dioxide; neutralisation
- (E) carbon dioxide; crystallisation
- (F) hydrogen; neutralisation
- (G) nitrogen; neutralisation

94. A bag has 10 balls numbered 1 to 10. One is drawn at random. Find the probability that its number is both even and greater than 5. [\[Data Handling\]](#)

- (A) $\frac{1}{2}$
- (B) $\frac{2}{10}$
- (C) 3
- (D) $\frac{3}{10}$
- (E) $\frac{4}{10}$
- (F) $\frac{5}{10}$
- (G) $\frac{7}{10}$

95. For a simple pendulum, the time period depends mainly on which property, rather than on how heavy the bob is? [\[Motion & Time\]](#)

- (A) the air around it
- (B) the bob's mass
- (C) the bob's colour
- (D) the size of the push given
- (E) the colour of the string
- (F) the time of day
- (G) its length

96. Find the value of the expression $3a - 2b$ when $a = 5$ and $b = 4$. [\[Algebraic Expressions\]](#)

- (A) 11
- (B) 17
- (C) -7
- (D) 23
- (E) 8
- (F) 7
- (G) 1

97. Distilled water is neutral. Of 90 water samples, $\frac{2}{9}$ were acidic and $\frac{1}{3}$ were basic, while the rest were neutral. How many were neutral, and what does neutral water do to litmus? [\[Acids, Bases & Salts\]](#)

- (A) 50 are neutral; no change to litmus
- (B) 40 are neutral; no change to litmus
- (C) 45 are neutral; no change to litmus
- (D) 40 are neutral; it turns litmus red
- (E) 30 are neutral; no change to litmus
- (F) 40 are neutral; it turns litmus blue
- (G) 20 are neutral; no change to litmus

98. Four cloth pieces measure 2.5 m, 3.0 m, 2.0 m, and 4.5 m. Find the mean length and the total length. [\[Data Handling\]](#)

- (A) mean 3 m; total 9 m
- (B) mean 12 m; total 3 m
- (C) mean 3 m; total 11 m
- (D) mean 4 m; total 12 m
- (E) mean 3 m; total 12 m
- (F) mean 3 m; total 12.5 m
- (G) mean 2.5 m; total 12 m

99. On three days a cyclist's speeds were 12, 18, and 15 km/h. Find her mean speed, and her total distance if she rode for 2 hours each day. [\[Motion & Time\]](#)

- (A) mean 15 km/h; 90 km total
- (B) mean 15 km/h; 30 km total
- (C) mean 16 km/h; 90 km total
- (D) mean 45 km/h; 90 km total
- (E) mean 14 km/h; 90 km total
- (F) mean 15 km/h; 45 km total
- (G) mean 15 km/h; 180 km total

100. Pens cost x rupees each and a notebook costs 20 rupees. Write the cost of 3 pens and 2 notebooks as an expression, and find its value when $x = 15$. [\[Algebraic Expressions\]](#)

- (A) $2x + 40$; Rs 70
- (B) $3x + 20$; Rs 65
- (C) $3x + 40$; Rs 75
- (D) $3x + 22$; Rs 67
- (E) $3x + 40$; Rs 85
- (F) $5x + 40$; Rs 115
- (G) $3x + 40$; Rs 45